

MOHIT RAJ

Machine Learning Engineer — Data Scientist — ML Research
mraj776384@gmail.com — +91-9315244734 — GitHub — LinkedIn

PROFESSIONAL SUMMARY

B.Tech (Hons.) Data Science graduate with hands-on experience in end-to-end Machine Learning pipelines, from data cleaning, EDA, and feature engineering to model training, hyperparameter tuning, and ensemble learning. Proficient in Python, Scikit-learn, XGBoost, Pandas, NumPy, and SQL, with ML research experience at NIT Karnataka.

TECHNICAL SKILLS

Programming	Python, SQL, C++
Machine Learning	Scikit-learn, XGBoost, Random Forest, Regression, Classification, Ensemble Learning, Feature Engineering, Feature Selection, Hyperparameter Tuning, Cross-Validation, Model Evaluation
Data Science	Pandas, NumPy, EDA, Data Cleaning, Data Preprocessing, Statistics, Data Visualization, Matplotlib, Seaborn
Deep Learning	TensorFlow, PyTorch
Tools & Platforms	Git, GitHub, Jupyter Notebook, VS Code, Google Earth Engine

PROFESSIONAL EXPERIENCE

AI & Robotics Coach — Stembotix

May 2026 – Present

- Delivered hands-on sessions on AI, Machine Learning, Robotics, and IoT, simplifying complex concepts through practical, project-based learning for diverse student cohorts.
- Guided students in developing Python and AI-based projects; strengthened technical communication and mentoring skills through regular workshop facilitation.

INTERNSHIP EXPERIENCE

Research Intern — Machine Learning & Remote Sensing — NIT Karnataka (NITK Surathkal)

June 2025

- Engineered an end-to-end crop classification ML pipeline on Sentinel-1 and Sentinel-2 satellite imagery using Google Earth Engine and Python, enabling scalable geospatial analysis for precision agriculture.
- Performed feature engineering by extracting vegetation indices (NDVI, EVI, OSAVI, RVI) from multispectral remote sensing data, directly improving predictive signal for downstream classification.
- Designed and evaluated a Random Forest classification model with systematic data preprocessing, hyperparameter tuning, and statistical validation on real-world agricultural datasets.

PROJECTS

House Price Prediction (Advanced Regression)

[GitHub](#)

Python · Pandas · NumPy · Scikit-learn · XGBoost · Ridge Regression · Gradient Boosting

- Built a full end-to-end regression pipeline: data cleaning, missing value imputation, outlier removal, and EDA; engineered features (TotalSF, TotalBathrooms, HouseAge); applied Box-Cox skewness correction, label encoding, one-hot encoding, and RobustScaler preprocessing.
- Trained Ridge, Gradient Boosting, and XGBoost with 5-fold cross-validation (RMSE) and RandomizedSearchCV tuning; constructed a weighted ensemble achieving the best CV RMSE across all candidates.

EDUCATION

B.Tech (Hons.) in Computer Science — Data Science — CGPA: 7.6/10

2022 – 2026

Chhattisgarh Swami Vivekanand Technical University

Intermediate (Class XII) — 70% — BSEB

2022

Secondary School (Class X) — 89.8% — CBSE

2020

CERTIFICATIONS

- Generative AI — RAG, AI Agents, Deployment (Udemy) — AI Dashboards using Microsoft Power BI